

From Talent Management to Agent Governance

What the Human Talent-Management Lifecycle Transfers to Multi-Agent AI, and Where the Analogy Breaks

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Organizations have spent roughly a century characterizing how to manage agents through a lifecycle of selection, socialization, performance management, and separation. This paper treats that lifecycle as a structured design lens for the governance of multi-agent AI systems. We review the management and organizational-psychology literature for each phase, drawing out the efficiency mechanism the phase provides and the failure mode it incurs, and we ask in each case what transfers to a population of software agents and what does not. The organizing observation is that the literature's strongest cautions, particularly its negative verdict on aggressive turnover and forced ranking, are driven by costs that are specific to human employment: lost human and social capital, turnover contagion, morale damage, tacit-knowledge loss, hiring expense, and legal exposure. These costs fall sharply in the multi-agent setting, where an agent can be archived and respawned at the cost of a few thousand tokens, though we are careful to show that they fall unevenly and that several do not reach zero. The literature therefore transfers not as a set of borrowed prescriptions but as a diagnostic that tells us which costs each lifecycle transition incurs and whether those costs are present for agents. The asymmetry's logical core is close to definitional, so the paper's weight rests not on it but on the boundary it exposes: the cost asymmetry holds only to the degree that an agent's knowledge is faithfully externalized in provenance records, since human socialization carries tacit knowledge that software provenance may not, and the same fact governs whether a principal can reliably identify which agents to separate at all. This reframes faithful reporting, which work on multi-agent systems often assumes, as the measurable binding constraint of the whole approach. We derive design implications for runtime governance layers, position them against existing proposals built on agent visibility and logging, and draw a hard normative line that we acknowledge is a commitment rather than a technical guarantee. This is a lens for governing software agents and for augmenting human oversight; it is not a prescription for managing human workers, for whom the same literature counsels the opposite.

Additional Key Words and Phrases: AI governance, multi-agent systems, talent management, principal-agent theory, organizational psychology, human-AI collaboration, automation and augmentation, runtime governance

1 Introduction

The manager and the AI orchestrator share a structural problem. Both are principals responsible for the performance of agents that operate beneath them, both must delegate to agents that hold local information, and both face the classic delegation hazard that agency theory describes [1, 6, 9, 15]. Both also rely, in the conventional case, on oversight that is fundamentally relational: a principal who attends to the agent, notices when it drifts, and intervenes. The difficulty with relational oversight is that it competes for a resource the principal holds in fixed supply. The attention-based view of organizations treats managerial attention as the scarce bottleneck through which activity must pass, allocated across competing demands under bounded capacity, so that attending more to one issue means attending less to others [21]. Oversight of any single agent is one such demand. Goal-disengagement theory sharpens what becomes of it under strain: an actor who perceives the gap between the current and desired state as too costly to close tends to disengage from that goal and reallocate effort rather than invest more [13]. Read together, with oversight as a goal the principal pursues, the two predict something uncomfortable: as load rises, attention is rationed and the oversight of any given agent is liable to be the effort that is withdrawn, precisely when the need for it is greatest. The conditions that generate the greatest need for oversight in multi-agent systems, high task volume, novel failure modes, and cascading delegation,

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are exactly the conditions that saturate a principal’s attention. This is the structural problem an AI governance layer must answer.

The move we make here is to treat the human talent-management lifecycle as a design lens. Organizations do not manage people through a single undifferentiated act; they move people through a sequence of phases, each with its own purpose and its own well-studied failure modes: they select who enters, they socialize newcomers into the role, they manage ongoing performance through feedback and evaluation, and they separate those who leave or are removed. The organizational-psychology and management literatures have spent decades characterizing the efficiency mechanism each phase provides and the way each phase fails. That accumulated characterization is a resource for AI governance, because a multi-agent system runs its agents through a structurally parallel sequence: it spawns agents, loads them with context, governs their behavior at runtime, and archives them at the end of their work. The question this paper asks, phase by phase, is what the human literature establishes, and which of those findings transfer to a population of software agents.

The organizing answer is that the transfer is selective in a specific and informative way. The management literature’s strongest cautions are about cost. The reason rigorous selection is worth its expense, the reason socialization is allowed to take months, the reason feedback is treated as a scarce managerial resource, and above all the reason the literature is so negative about high turnover and forced ranking, is that each of these transitions is costly and difficult to reverse for human employees. A bad hire is expensive to undo. A departure destroys human and social capital, can trigger further departures, and carries the firm’s tacit knowledge out the door. Those costs are the engine behind the prescriptions. In the multi-agent setting the costs fall sharply, though as we will show not uniformly to zero. The literature therefore does not transfer as a set of prescriptions to copy. It transfers as a diagnostic: for each phase, it tells us which cost the phase is built to manage, and we then ask whether that cost is present for agents. Where the cost is absent, the design space opens; where it reappears, the analogy breaks and the human caution applies again.

We state our normative position before developing the argument, because the framing matters and is easy to misread. The claim that aggressive separation of underperforming *agents* is justified must not be heard as a claim that organizations should treat human workers that way. The opposite is true, and the same literature we draw on says so. The relationship between automation and augmentation is not a clean choice between them: Raisch and Krakowski argue that the two are interdependent, that over-emphasizing either produces self-reinforcing negative cycles, and that the durable position holds both together [25]. Our framing is precise about where each falls. A structural governance layer *automates* the operational component of oversight, the moment-to-moment monitoring and correction that a manager performs in real time, while it *augments* the principal’s strategic oversight, the human work of setting policy and reviewing what the agents did. This is the entanglement the paradox describes rather than an escape from it, and it is consistent with the human-automation literature’s insistence that the question is not whether to automate but which functions to automate and at what level [22]. The human principal is therefore a standing requirement of the design, not a cost to be removed. We return to the algorithmic-management literature that polices this line [17] as an explicit boundary in Section 9.

Contributions. (1) We introduce the talent-management lifecycle as a structured lens for reasoning about agentic AI governance, and review the literature of each phase (Sections 2–6). (2) We identify a cost-structure asymmetry as the principle that organizes what transfers, while acknowledging that its logical core is close to definitional, which is why the paper’s weight rests not on the asymmetry but on the boundary it exposes: that faithful provenance is the measurable constraint on which cheap, structural governance depends (Sections 7 and 9). (3) We derive design

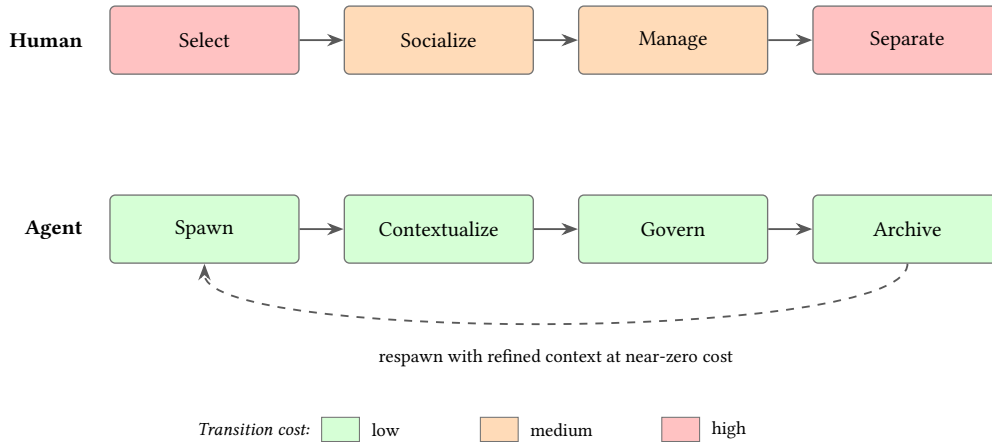


Fig. 1. The human and agent lifecycles aligned by phase. Selection corresponds to spawning and configuration, socialization to context loading and inheritance, performance management to runtime governance, and separation to archival. In the human lifecycle, entry and exit are the dominant cost drivers and exit is effectively irreversible, which constrains the principal's response options under load. In the agent lifecycle, every transition is low-cost and the dashed loop is the option unavailable in human management: when an agent underperforms, archive it and respawn with refined context. The asymmetry the figure depicts is the organizing device of the paper, not by itself its contribution.

implications for runtime governance layers, the central one being to treat provenance fidelity as a first-class, measured property, and we position these against existing agent-governance proposals built on visibility and logging (Section 8). (4) We state the boundary conditions under which the analogy breaks, including the provenance-fidelity boundary and a normative boundary that separates agent governance from human workforce management (Section 9).

2 The Talent-Management Lifecycle as a Lens

We use a deliberately simplified four-phase model of the employee lifecycle. Real talent management includes recruitment, development, succession, and compensation design, among others; we collapse the lifecycle to the four phases that have the cleanest agent analogues and the deepest empirical literatures. We set development aside in particular, even though it has a clear agent counterpart in fine-tuning, tool acquisition, and retrieval augmentation, because development raises an agent's underlying capability while our analysis takes capability as given; we return to that boundary in Section 9. *Selection* is the decision about who enters. *Socialization* is the process by which a newcomer becomes a functioning member of the unit. *Performance management* is the ongoing work of directing and evaluating an incumbent, of which feedback is the central instrument. *Separation* is the removal or departure of a member, whether voluntary or imposed.

Each phase has a structural counterpart in the agent lifecycle. Selection corresponds to the choice of which model, configuration, and mandate to instantiate. Socialization corresponds to loading the agent with context, including the inherited record of prior agents' work. Performance management corresponds to runtime policy enforcement, the always-available substitute for a manager's intermittent feedback. Separation corresponds to archival and, where warranted, termination and respawn. Figure 1 sets the two lifecycles side by side and previews the cost asymmetry that the rest of the paper develops: in the human lifecycle, the two transitions that bound the relationship, entry and exit, are the dominant cost drivers and constrain the principal's options; in the agent lifecycle, every transition is low-cost and exit is reversible.

The lens is framework-agnostic. It does not depend on any particular orchestration substrate; it is a way of importing a century of accumulated knowledge about managing agents under cost and irreversibility, and of asking which of that knowledge survives when the costs change. We draw out design implications for runtime governance layers in Section 8, but the analysis in Sections 3–7 stands independent of any specific system.

Relation to the human-automation and human-AI collaboration literatures. Posing governance as a question of which oversight functions to encode and which to leave to a human principal places this work alongside the human-automation literature, which distinguishes types and levels of automation across the functions of information acquisition, information analysis, decision selection, and action implementation, and warns that automation does not merely supplant human activity but reshapes it and can impose new coordination demands on the operator [22]. In those terms, the lens we develop is a claim about which of these functions a governance layer should automate for an agent population and which it should preserve for the human principal, a question we return to in Sections 5 and 9. The work also sits within computer-supported cooperative work and human-AI collaboration, which study how teams coordinate through the digital traces that online work produces and how to design systems that support rather than displace human judgment [10, 25]. Our reliance on provenance as the substrate for governance is an instance of that digital-trace tradition: the record an agent leaves is at once the audit trail the principal reviews and the inheritance the next agent reads.

Relation to work on multi-agent systems and agent governance. Multi-agent LLM systems, in which several model instances are orchestrated to decompose and execute tasks, are now widely studied and built [24, 37, 38], and a growing literature asks how to govern them. Two strands are closest to this paper. The first is empirical: a recent failure taxonomy built from more than a thousand annotated traces across seven multi-agent frameworks finds that the dominant failure modes are not raw capability gaps but specification and role-clarity problems, inter-agent misalignment, and inadequate verification of outputs [7]. The second is prescriptive: work on visibility into AI agents proposes agent identifiers, real-time monitoring, and activity logging as the mechanisms through which oversight and accountability are achieved [8], and related proposals sketch governance frameworks for agentic systems [26]. The lifecycle lens is complementary to both. Where the visibility literature establishes *what* to record and the failure taxonomy establishes *which* failures occur, the lens supplies a *why*: by importing the tacit-knowledge problem from the human socialization literature, it identifies the fidelity of the recorded provenance, rather than its mere existence, as the property that bounds how cheaply agents can be governed and replaced, and it ties that property to the specific failure modes the empirical work observes.

3 Phase 1: Selection

Efficiency. Selection is the most heavily studied phase of the human lifecycle, and for a clear reason: it is the highest-leverage decision the principal makes, because its consequences are sticky. The long-standing synthesis is Schmidt and Hunter’s meta-analytic review of eighty-five years of personnel-selection research [29, 30], which positioned general mental ability as the strongest single predictor of job performance, with corrected validity near .5 and composites reaching the low .60s. Those specific figures should now be treated with care. A recent re-analysis showed that the range-restriction corrections underlying decades of meta-analyses were systematically too large, that operational validities are substantially lower than the canonical numbers, with general mental ability falling to roughly .31, and that structured interviews, at roughly .42, displace it as the highest-validity single predictor [27]. What survives the revision, and what the lens actually relies on, is the qualitative claim rather than the coefficients: selection is predictively valid, it

is costly to do well, and its consequences are sticky, so getting it right or wrong compounds over the entire tenure of the hire.

Inefficiency. The same literature documents the limits. Even the best methods leave most of the variance in performance unexplained, so selection is a probabilistic bet rather than a guarantee. The most studied predictor, general mental ability, carries well-known adverse-impact and fairness concerns when used as a gate. Rigorous selection is slow and expensive. And the deepest inefficiency is structural rather than methodological: human selection is close to a one-shot decision. The cost of reversing a hire is high enough that the firm is forced to invest heavily in predicting performance in advance, because it cannot cheaply learn by trying.

What transfers. The agent analogue of selection is the choice of which model, configuration, and prompt to instantiate for a mandate. The selection lens predicts a capability floor: there is a level of raw competence below which no amount of structural guidance compensates, because the binding constraint becomes the agent's ability rather than its governance. This is the agent counterpart of the observation that no management practice repairs a fundamental capability gap, and it implies that the first selection decision, choosing a model above the floor, is real and consequential even though everything downstream of it is cheap.

Where the analogy breaks. The elaborate predictive apparatus that human selection requires exists because human selection is sticky and expensive. For agents, that premise is false. A configuration can be instantiated, evaluated against the task, and discarded in favor of another at trivial cost, and the decision is fully reversible. The optimal agent-selection strategy is therefore not better prediction but cheap empirical iteration: try a configuration, observe, and reconfigure. The human inefficiency, the obligation to predict because one cannot cheaply learn, is precisely the inefficiency the agent setting removes. The validity-maximizing machinery of personnel selection is an answer to a problem that agents do not have.

4 Phase 2: Socialization

Efficiency. Once a newcomer is hired, the organization must turn them into a functioning member of the unit. The meta-analytic anchor is Bauer and colleagues [4], who model newcomer adjustment as a mediator: organizational socialization tactics and the newcomer's own information seeking improve three adjustment outcomes, role clarity, self-efficacy, and social acceptance, and those in turn drive performance, commitment, intention to remain, and reduced turnover. Role clarity is the load-bearing construct, and it is the same construct that the feedback and role-ambiguity literatures place at the center of performance [11, 35]. Socialization is, in large part, the process of resolving a newcomer's uncertainty about what the role requires.

Inefficiency. Socialization is slow, relational, and lossy. Adjustment unfolds over months, during which the newcomer is below full productivity. Much of what is transferred is tacit, conveyed through observation and interaction rather than written down, which makes it both slow to transmit and vulnerable to loss when the people who hold it leave. And the process can be actively undermined: where organizational insiders undermine rather than support the newcomer, adjustment outcomes are substantially worse.

What transfers. The agent analogue of socialization is the loading of context, including the inherited record of prior agents' work, and role clarity maps directly onto a well-specified mandate. The socialization literature's central finding, that role clarity mediates the path from onboarding to performance [4], yields a concrete prediction for agent systems:

the precision of an agent’s mandate should govern how decisively it can act, with an underspecified mandate forcing the agent to guess or to seek clarification and a precise one enabling single-pass execution. That prediction is no longer purely speculative. The largest single category in a recent empirical taxonomy of multi-agent failures is specification and role-clarity problems, agents misinterpreting or under-adhering to their task and role definitions [7], which is the agent-systems echo of the role-clarity finding. What remains untested is the directional and quantitative form of the relationship, which a controlled manipulation of mandate precision could establish.

Where the analogy breaks. The time cost and the relational component of human socialization vanish for agents. Context loads in one step rather than over months, and social acceptance is moot, since an agent has no peers whose acceptance it needs in order to function. But this phase also surfaces the first appearance of the boundary that bounds the whole argument. Human socialization transfers tacit knowledge that is never explicitly recorded; human memory and relationships carry that load silently. For agents there is no silent carrier. Whatever an agent knows must be externalized in provenance and inheritance records, or it is lost when the agent is archived. To the degree those records are faithful, agent socialization is strictly cheaper and faster than the human version. To the degree they are lossy, cheap respawn re-incurs a cold-start cost on every cycle, and the asymmetry weakens. We develop this in Section 9.

5 Phase 3: Performance Management and Feedback

Efficiency. The central instrument of ongoing performance management is feedback, and its functions are well established. Feedback reduces role ambiguity, improves task performance, and signals the principal’s investment in the agent [2, 11, 16, 18, 33]. These functions are the day-to-day mechanism by which a principal keeps an incumbent aligned with expectations after selection and socialization are complete.

Inefficiency. Feedback has two documented weaknesses. First, it is unreliable even when delivered: the foundational meta-analysis on feedback interventions finds that a substantial share of them actually reduce performance, depending on where the feedback directs attention [18]. Second, and more consequentially for governance, feedback is relational and availability-dependent, and its availability is governed by where the principal’s scarce attention is allocated [21]. The self-regulation literature describes how individuals allocate finite effort across competing demands by the magnitude of perceived discrepancies [19, 20, 28, 36], and the goal-disengagement mechanism predicts that when a discrepancy is perceived as too costly to close, the response is not to allocate more effort but to disengage and reallocate elsewhere [13]. Applied to a principal under load, these together predict that feedback effort falls as perceived difficulty and competing demand rise, which is to say the instrument is least available when it is most needed. Even a conservative reading that treats feedback as merely intermittent rather than actively withdrawn is sufficient for the argument that follows.

What transfers. Structurally encoded policy performs the same role as feedback, communicating expectations and constraining behavior toward desired outcomes, but it operates independent of principal availability and applies uniformly to every action rather than intermittently when the principal happens to attend. This feedback-policy equivalence is the conceptual hinge of the lens: the function feedback performs relationally in human organizations, policy performs structurally in agent systems. The content of the structural signal matters, and there is evidence for this from a neighboring area: constitutional approaches to model alignment show that natural-language principles produce measurable behavioral shifts [3], which suggests that the specific content of an agent’s encoded policy, not merely its presence, is causally connected to behavior.

Where the analogy breaks. This is the phase where the asymmetry most cleanly favors the agent case, and it does so by removing an inefficiency rather than a cost. The defining weakness of human feedback is that it is available only when the principal is available, and on the weaker reading it is at least intermittent. Structural policy has no such dependency: it is always on. In the terms of Section 2, this is the operational oversight function that a governance layer automates, converting intermittent, availability-bound feedback into a constant property of the system.

6 Phase 4: Separation

Separation is where the cost asymmetry is sharpest, where the question of high-turnover strategies belongs, and where the framing risk is greatest. We treat it at more length than the other phases.

The aggressive-separation hypothesis. A strand of practitioner thought holds that aggressive separation of weaker performers raises organizational performance. Its most prominent recent statement is the Netflix model described by Hastings and Meyer [12], built on the premise that one excellent contributor is worth more than several adequate ones and that adequate performers impose coordination costs on excellent ones, with the so-called keeper test as the dismissal mechanism and high talent density as the goal. An older institutional form of the same intuition is the forced-distribution rating system, or rank-and-yank, popularized by General Electric, whose proponents have cited a simulation showing that forced ranking with dismissal improves a firm's modeled workforce potential over an initial period [31].

What the evidence actually shows. The peer-reviewed evidence does not support aggressive separation as a general performance strategy, and on balance points the other way. The meta-analytic relationship between turnover and organizational performance is significant and negative, and it is more negative for voluntary and reduction-in-force turnover than for involuntary turnover [23]. The relationship is curvilinear, and turnover has a contagion quality in which one departure raises the likelihood of others. The forced-ranking field evidence is sharper still. In a multinational firm that enforced a strict ranking cutoff, downgraded employees were at least thirty-four percent more likely to leave voluntarily, and the effect persisted even though the firm avoided calibration bias, reassured employees about their career prospects, and paid the downgraded employees larger bonuses than their top-ranked peers [5]. The institutional history reinforces the point: the practice generated sustained legal exposure and was abandoned by several of its most prominent adopters. The Netflix case, by contrast, is a single firm's self-narrated account; its own careful readers note that copying the practices of one successful, path-dependent organization confuses correlation with causation. It can motivate a hypothesis. It cannot establish that the strategy causes performance.

The reframe. The decisive observation is not that aggressive separation is good or bad in the abstract, but *why* it is bad for human organizations. The negative verdict is driven by costs, and the costs are specific to human employment. A departure destroys firm-specific human capital and the social capital embedded in the departed person's relationships. It is contagious, raising the probability of further departures. It damages the morale of those who remain. It carries tacit knowledge out of the firm, because human knowledge is not fully externalized. It incurs the expense of selecting and socializing a replacement. It exposes the firm to legal risk. And forced ranking in particular introduces a perverse incentive structure, in which employees competing for a fixed quota of top ratings are rewarded for hoarding knowledge and for not developing their peers, the opposite of what the organization wants. Each of these is a cost, and each is a reason restraint is the right policy for human separation.

Now apply the diagnostic, taking care not to overstate the result. Several of those costs genuinely do approach zero. Archiving and respawning an agent destroys no human capital and no interpersonal relationships, damages no morale, and creates none of the ranking-driven incentive distortion that forced ranking imposes on people, because agents do not compete for a quota and have no reputational stake to defend, and so have no reason to hoard knowledge or withhold help. Recruiting and onboarding expense collapses to a few thousand tokens. Three of the costs, however, fall but do not vanish, and the honest version of the argument names them. First, separation is not fully non-contagious: there is no morale contagion, but respawning an agent that other agents depended on can disrupt those downstream, a coordination failure of exactly the kind that empirical studies find to be among the most common in multi-agent systems [7], and the event leaves traces in the registry and provenance that the system must reconcile. Second, the tacit-knowledge cost is only discharged to the degree the agent’s knowledge was externalized; what was not written into provenance leaves with the agent, exactly as it does for a departing employee, and coordination knowledge that is a property of the agent population rather than of any single agent may not reduce cleanly to per-agent records. Third, legal and accountability exposure does not disappear; the agent has no wrongful-dismissal or labor claim, but the operator remains accountable for what the agents do, and that exposure is if anything heightened by a policy of rapid, lightly checked turnover. The accurate claim is therefore not that agent separation is costless, but that its cost structure is different and largely lower, with the residual costs concentrated in coordination and in the fidelity of the provenance record.

The conclusion, and the firewall. It follows that an AI-native separation policy can be steeper and faster than human management would tolerate, not because human management is too cautious, but because the cost structure that justifies its caution is largely absent for software. This is the point at which the augmentation framing is load-bearing rather than decorative. We are not claiming that the Netflix model is validated, because the evidence says it is not. We are not claiming that organizations should separate human workers more aggressively, because the same literature we rely on counsels the opposite, and the harms it documents, exit of high performers, contagion, and knowledge-hoarding incentives, are real human harms. We are making a narrow claim about one governance practice applied to one kind of entity: aggressive termination of an underperforming *software agent* is licensed by a cost structure largely unique to software. Importing the human-management caution onto agents would be a category error, and exporting the agent policy onto humans would be exactly the harm the algorithmic-management literature documents. Both directions are errors, and Section 9 draws the line between them explicitly.

Two qualifications. Permissibility is not optimality. A policy of steep, fast termination is licensed by the cost structure, but it is not automatically the best policy, because termination forfeits any work in progress that respawn does not reliably inherit, trading a known cost, lost progress, against an uncertain one, continued underperformance. More fundamentally, the case for aggressive separation presupposes that the principal can identify which agents to separate. Part of why human forced ranking misfires is a measurement failure: most of the variance in performance ratings is idiosyncratic to the rater rather than attributable to the ratee [32], so the system cannot reliably find its bottom performers in the first place. If agents’ self-reports are themselves unreliable, as both the literature on strategic underperformance and evaluation gaming [14, 34] and the empirical prevalence of verification failures in multi-agent systems [7] suggest they may be, then the same misclassification risk applies to agents, and steep separation may be acting on noise. This is where Section 6 hands off to the boundary conditions of Section 9.

Phase	Human cost driver (why the practice exists)	Agent cost	Governance implication
Selection	Hiring is sticky and expensive, so performance must be predicted in advance	Near-zero, reversible	Prefer cheap empirical iteration over predictive validation; gate only on the capability floor
Socialization	Adjustment is slow and tacit knowledge is lossy	Near-zero <i>if</i> provenance is faithful	Specify mandates precisely; externalize knowledge in provenance rather than relying on tacit transfer
Performance mgmt.	Feedback is scarce and at best intermittent under principal load	None (always-on)	Encode policy structurally so oversight is invariant to principal availability
Separation	Exit destroys capital, is contagious, and is legally and morally costly	Lower, not zero (operator liability; dependency disruption)	Permit steeper but not unconditional sanctions; preserve in provenance what an exit interview would lose, and check the signal before acting

Table 1. The cost-structure asymmetry across the lifecycle. The human literature’s prescriptions are responses to the cost and irreversibility of each transition. Those costs are absent, conditional, or merely lower for agents, which is what determines whether each human prescription transfers.

7 The Cost-Structure Asymmetry

The four phases share a single underlying pattern. In each, the human literature’s prescription is shaped by the cost and the irreversibility of the corresponding transition. Selection is studied intensively because hiring is sticky. Socialization is allowed to be slow because the alternative, rapid turnover, is costly. Feedback is treated as a scarce resource because the principal’s attention is finite. Separation is approached with restraint because exit is expensive and irreversible. The prescriptions are rational responses to a cost structure.

The multi-agent setting changes the cost structure, and it does so in the same direction across all four phases: it lowers transition costs and makes transitions reversible. This is why the lifecycle lens transfers as a diagnostic rather than as a set of borrowed prescriptions. For each phase we ask which cost the human practice is built to manage, and whether that cost is present for agents. Table 1 records the result. The general design heuristic that falls out is the inverse of the human one: where the human literature invests in expensive prediction and relational maintenance to cope with irreversibility, the agent setting should prefer cheap reversible iteration, that is, respawn, reconfigure, and re-prompt.

What the asymmetry does and does not establish. Stated abstractly, the asymmetry is close to definitional: if a practice exists to manage a cost, and that cost is absent, the practice is not needed. The work is therefore not done by the abstract claim but by the two questions it raises. The first is whether the costs really are absent for agents, which Section 6 argues is mostly but not entirely true. The second, and the one that carries the paper’s weight, is where the costs reappear. The lens earns its keep precisely here: by importing the tacit-knowledge problem from the socialization literature and the rating-noise problem from the performance-measurement literature, it predicts that the binding constraint on cheap-respawn governance is the fidelity of the provenance record, a constraint that does not fall out of a purely systems-oriented framing on its own. A reframing that only restated that software is cheaper to replace than people would not be worth the apparatus. A reframing that locates a specific, testable failure boundary is. That boundary, faithful provenance as the measurable constraint on cheap governance, is the paper’s central claim, and the rest of the paper states it precisely and draws out the design and empirical commitments it implies.

8 Design Implications for Runtime Governance Layers

The lifecycle analysis yields a set of requirements that any runtime governance layer for a multi-agent system should satisfy, and it yields them by derivation rather than assertion. From selection: the layer should gate on the capability floor, since below it no governance produces lift, and otherwise treat model and configuration choice as cheap reversible experiments rather than high-stakes predictions. From socialization: the layer should make mandate specification precise enough to deliver role clarity, and should externalize agent knowledge in a provenance record that substitutes for the tacit transfer human socialization provides. From performance management: the layer should encode policy structurally and validate actions before execution, so that operational oversight is a constant property of the system rather than a function of principal availability. From separation: the layer should permit graduated but steeper sanctions, including cheap termination and respawn, while ensuring that what a departing agent knew is preserved in provenance, so that respawn inherits rather than restarts, and while checking the triggering signal before acting on it.

The most consequential of these implications is not a feature but a constraint, and it is the lens's distinctive prediction. Proposals for visibility into AI agents already establish the mechanisms through which a system can record agent behavior: agent identifiers, real-time monitoring, and activity logs [8]. What the lifecycle lens adds is a claim about which property of that record matters most, and why: not the existence of logs but their *fidelity*, because the cheapness of agent separation, the value of inheritance, and the trustworthiness of the signal that triggers sanctions all depend on it, and because the dominant empirical failures of multi-agent systems include precisely the verification and specification gaps that lossy or unfaithful records would produce [7]. The design implication is therefore to treat provenance fidelity as a first-class, measured property of a governance layer rather than an assumed one. This is a requirement the lens generates, that a purely architectural framing may not, and that complements rather than competes with the visibility mechanisms the governance literature has proposed.

9 Boundary Conditions: Where the Analogy Breaks

The lens is productive precisely because it identifies a shared structural problem across two domains. It is less productive when it papers over the places where the domains genuinely differ. We state the boundaries explicitly, because they are as much a contribution as the transfers.

9.1 The provenance-fidelity boundary

The entire cost asymmetry rests on one assumption: that a departing agent's knowledge is faithfully externalized, so that archival and respawn lose nothing of value. This is the agent counterpart of the tacit-knowledge problem in human socialization, and it is the boundary that bounds the whole argument. Human organizations carry tacit knowledge silently, in memory and relationships, which is why human separation is so costly and why the literature counsels restraint. Agent populations have no silent carrier; they have only the provenance record. If that record is faithful, the asymmetry holds and agent separation is genuinely cheap. If it is lossy, cheap respawn re-incurs a cold-start cost on every cycle, and the agent case begins to resemble the human one it claimed to escape.

Two open questions sit on this boundary. The first is whether agents report faithfully at all. There is emerging evidence that language models can strategically underperform on evaluations and can behave deceptively in ways that persist through training [14, 34], which makes motivated or inaccurate self-reporting a live concern rather than a hypothetical one. Independent of any intent to deceive, large-scale analysis of multi-agent systems finds inadequate verification of outputs to be one of the three dominant failure categories [7], so unreliable self-assessment is a documented property of

current systems and not merely a theoretical risk. The connection to Section 6 is direct: human forced ranking misfires in part because performance measurement is noise-dominated, with most rating variance attributable to the rater rather than the ratee [32], and an agent population whose self-reports are unreliable inherits the same measurement failure. The cheap-separation argument and the provenance-fidelity boundary are therefore two sides of one fact: separation is cheap only when the record is trustworthy, and the record is what tells the principal whom to separate. The second open question is whether natural-language policy is interpreted with low variance across runs, since a policy that is read differently each time cannot be relied on to externalize expectations consistently. Both questions are empirical, and both must be answered before the cost asymmetry can be asserted as established rather than argued.

9.2 The single-principal boundary

The analysis models a single coherent principal who sets policy and reviews provenance. Human talent management is inherently multi-stakeholder: the worker, the manager, the human-resources function, and peers all hold interests, and the lifecycle is negotiated among them. A multi-principal extension of the lens, in which several principals with divergent objectives govern a shared agent population, is unmodeled here and is a natural direction for further work.

9.3 The normative boundary

The sharpest boundary is normative, and we engage it directly because the lens borrows the vocabulary of talent management and that vocabulary carries risk. Kellogg, Valentine, and Christin characterize algorithmic control in the workplace as a contested terrain of power between employers and workers, operating through mechanisms they group as restricting, recording, rating, replacing, and rewarding [17]. The mechanisms this paper develops for governing software agents, total behavioral recording through provenance, pre-execution restriction, and immediate replacement, are recognizably members of that family. The risk is therefore concrete: governance mechanisms designed for software could be read as, or repurposed as, control mechanisms for human workers, which is the precise harm that literature documents.

We draw a hard line, and we are explicit that it is a normative commitment rather than a technical guarantee. The practices justified in this paper are justified *only because* the governed entity is software, and that justification is conditional on a premise we state rather than assume: that current software agents lack the morally relevant properties, a livelihood, a stake in their own continuation, a dignity interest, that make these same practices harmful when applied to people. We recognize that the moral status of AI systems is contested, and that a fairness-and-accountability audience includes researchers who work on exactly that question; our argument does not require the premise to be settled, only that the cost asymmetry holds for the entities actually being governed. Two honest limitations follow. First, the lens does not itself prevent its own misuse: control tooling has a long history of migrating from the domain where it is appropriate to the domain where it is not, regardless of the designer’s intent, and nothing in the analysis stops agent-grade surveillance and disposability from being pointed at human workers, which would be the harm the algorithmic-management literature warns against and which the evidence in Section 6 shows does not even achieve its stated performance goal for humans. The visibility literature itself notes that more detailed recording imposes escalating intrusiveness and privacy costs [8], costs that are acceptable for software and corrosive for people. Second, the augmentation framing constrains but does not eliminate the automation it contains. Consistent with the paradox [25], the layer automates operational oversight while augmenting strategic oversight, and the human-automation literature warns that a principal who only reviews provenance after the fact occupies the monitoring role most prone

to losing situational awareness [22]. Keeping the human principal genuinely in the loop, rather than nominally atop an audit trail they do not read, is therefore a design obligation rather than a solved problem.

9.4 The capability-threshold caveat

Finally, the entire analysis presupposes a model above the capability floor and holds the agent’s capability fixed. Raising capability is the work of what a fuller lifecycle would call development, whose agent analogue is fine-tuning, tool acquisition, and retrieval augmentation, and which we have deliberately bracketed. Below the floor, governance produces no lift, because the limiting factor is raw capability rather than management. The lens describes how to govern a capable agent well; it does not describe how to make an incapable one capable, and it says nothing about the development phase that would.

10 Limitations and Future Work

The transfer claims in this paper are conceptual, and the cost asymmetry that organizes them is argued rather than measured. The most valuable next step is to measure the boundary the paper rests on, and a direct protocol is feasible with existing infrastructure. Take a task that requires accumulated context, run an agent to a checkpoint, archive it, and respawn a successor whose only inheritance is the provenance record rather than the predecessor’s full working state; the difference in task performance between the full-state condition and the provenance-only condition is a direct estimate of the information lost across the archive-and-respawn boundary, and therefore of provenance fidelity. A near-zero gap would corroborate the cost asymmetry; a large gap would show that cheap respawn re-incurs a cold-start cost and would bound the claim. Two further studies follow from the analysis: a controlled manipulation of mandate precision, testing the socialization prediction of Section 4 that precision governs decisiveness, and an audit of agent self-reports against independent records of what occurred, testing the reporting-fidelity question of Section 9. The four-phase lifecycle model is itself a simplification of a richer human process that also includes recruitment, development, compensation, and succession, and extending the lens to those phases, development in particular, may surface additional transfers and additional boundaries. Finally, the cross-domain framing rests on an open question: whether the attention-allocation and goal-disengagement mechanisms that the organizational literature documents in human principals apply with the same force when the principal is itself an AI orchestrator, or when the cost of agent replacement is near-zero. The structural claim that relational oversight degrades under load survives that uncertainty; the specific mechanism may not.

11 Conclusion

The human talent-management literature is, read at the right level of abstraction, a long study of how to manage agents under cost and irreversibility, and its value for AI governance is the diagnosis it provides: phase by phase, it names the cost each transition incurs and the failure each managerial practice exists to prevent. The multi-agent setting lowers most of those costs and makes transitions reversible, and that change determines what transfers. Where a human practice exists to manage a cost that agents largely do not incur, the practice does not carry over and the design space opens: selection becomes cheap iteration, feedback becomes always-on policy, and separation can be steeper than human management would tolerate. Where a cost reappears, the analogy breaks, and the central reappearance is the tacit-knowledge problem, which for agents becomes a dependency on faithful provenance that human memory and relationships used to discharge silently.

The contribution is the lens and, above all, the boundary it exposes. The cost-asymmetry principle organizes what transfers, but it is close to definitional, so the paper's weight rests on a single load-bearing claim: that the value of cheap, structural agent governance is bounded by the fidelity of the provenance record, which reframes faithful reporting from an assumption the systems literature makes into a property that must be measured. The design implications, and the empirical studies that would test them, follow from that claim. Throughout, the direction of the argument is fixed: this is a method for governing software agents and for extending human oversight over them, and it is explicitly not a method for managing human workers, for whom the same literature, consulted honestly, counsels the opposite.

References

- [1] George A. Akerlof. 1970. The market for "lemons": Quality uncertainty and the market mechanism. *The Quarterly Journal of Economics* 84, 3 (1970), 488–500.
- [2] Susan J. Ashford and Anne S. Tsui. 1991. Self-regulation for managerial effectiveness: The role of active feedback seeking. *Academy of Management Journal* 34, 2 (1991), 251–280.
- [3] Yuntao Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, Jackson Kernion, et al. 2022. Constitutional AI: Harmlessness from AI Feedback. arXiv:2212.08073.
- [4] Talya N. Bauer, Todd Bodner, Berrin Erdogan, Donald M. Truxillo, and Jennifer S. Tucker. 2007. Newcomer adjustment during organizational socialization: A meta-analytic review of antecedents, outcomes, and methods. *Journal of Applied Psychology* 92, 3 (2007), 707–721.
- [5] Brittany Bond. 2025. Cut to the Curve: Underrecognition and Talent Loss from Forced Ranking in a Multinational Firm. *Management Science* (2025). Advance online publication. <https://doi.org/10.1287/mnsc.2023.01204>
- [6] Nick Bostrom. 2014. *Superintelligence: Paths, Dangers, Strategies*. Oxford University Press, Oxford, UK.
- [7] Mert Cemri, Melissa Z. Pan, Shuyi Yang, Lakshya A. Agrawal, Bhavya Chopra, et al. 2025. Why Do Multi-Agent LLM Systems Fail? In *Advances in Neural Information Processing Systems 38 (NeurIPS 2025), Datasets and Benchmarks Track*. Also arXiv:2503.13657.
- [8] Alan Chan, Carson Ezell, Max Kaufmann, Kevin Wei, Lewis Hammond, Herbie Bradley, Emma Bluemke, Nitarshan Rajkumar, David Krueger, Noam Kolt, Lennart Heim, and Markus Anderljung. 2024. Visibility into AI Agents. In *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency (FAccT '24)*. ACM, New York, NY, 958–973. <https://doi.org/10.1145/3630106.3658948>
- [9] Kathleen M. Eisenhardt. 1989. Agency theory: An assessment and review. *Academy of Management Review* 14, 1 (1989), 57–74.
- [10] Sharon A. Ferguson, Georgia Zande, and Alison Olechowski. 2024. No risk, no reward: Towards an automated measure of psychological safety from online communication. In *Extended Abstracts of the 2024 CHI Conference on Human Factors in Computing Systems (CHI EA '24)*. ACM, New York, NY. <https://doi.org/10.1145/3613905.3650923>
- [11] Cynthia D. Fisher and Richard Gitelson. 1983. A meta-analysis of the correlates of role conflict and ambiguity. *Journal of Applied Psychology* 68, 2 (1983), 320–333.
- [12] Reed Hastings and Erin Meyer. 2020. *No Rules Rules: Netflix and the Culture of Reinvention*. Penguin Press, New York, NY.
- [13] John R. Hollenbeck, Charles R. Williams, and Howard J. Klein. 1989. An empirical examination of the antecedents of commitment to difficult goals. *Journal of Applied Psychology* 74, 1 (1989), 18–23.
- [14] Evan Hubinger, Carson Denison, Jesse Mu, Mike Lambert, Meg Tong, et al. 2024. Sleeper Agents: Training Deceptive LLMs that Persist Through Safety Training. arXiv:2401.05566.
- [15] Michael C. Jensen and William H. Meckling. 1976. Theory of the firm: Managerial behavior, agency costs and ownership structure. *Journal of Financial Economics* 3, 4 (1976), 305–360.
- [16] Jaehee Jong. 2016. The role of performance feedback and job autonomy in mitigating the negative effect of role ambiguity on employee satisfaction. *Public Performance and Management Review* 39, 4 (2016), 814–834.
- [17] Katherine C. Kellogg, Melissa A. Valentine, and Angèle Christin. 2020. Algorithms at Work: The New Contested Terrain of Control. *Academy of Management Annals* 14, 1 (2020), 366–410.
- [18] Avraham N. Kluger and Angelo DeNisi. 1996. The effects of feedback interventions on performance: A historical review, a meta-analysis, and a preliminary feedback intervention theory. *Psychological Bulletin* 119, 2 (1996), 254–284.
- [19] Robert G. Lord, James M. Diefendorff, Aaron M. Schmidt, and Rosalie J. Hall. 2010. Self-regulation at work. *Annual Review of Psychology* 61 (2010), 543–568.
- [20] Andrew Neal, Timothy Ballard, and Jeffrey B. Vancouver. 2017. Dynamic self-regulation and multiple-goal pursuit. *Annual Review of Organizational Psychology and Organizational Behavior* 4 (2017), 401–423.
- [21] William Ocasio. 1997. Towards an attention-based view of the firm. *Strategic Management Journal* 18, S1 (1997), 187–206.
- [22] Raja Parasuraman, Thomas B. Sheridan, and Christopher D. Wickens. 2000. A model for types and levels of human interaction with automation. *IEEE Transactions on Systems, Man, and Cybernetics - Part A: Systems and Humans* 30, 3 (2000), 286–297.
- [23] Tae-Youn Park and Jason D. Shaw. 2013. Turnover rates and organizational performance: A meta-analysis. *Journal of Applied Psychology* 98, 2 (2013), 268–309.

- [24] Joon Sung Park, Joseph C. O'Brien, Carrie J. Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. 2023. Generative Agents: Interactive Simulacra of Human Behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST '23)*. ACM, New York, NY.
- [25] Sebastian Raisch and Sebastian Krakowski. 2021. Artificial Intelligence and Management: The Automation-Augmentation Paradox. *Academy of Management Review* 46, 1 (2021), 192–210.
- [26] Muhammad H. Raza, et al. 2025. A governance framework for agentic AI: Mitigating systemic risks in LLM-powered multi-agent architectures. *Journal of International Crisis and Risk Communication Research* (2025).
- [27] Paul R. Sackett, Charlene Zhang, Christopher M. Berry, and Filip Lievens. 2022. Revisiting meta-analytic estimates of validity in personnel selection: Addressing systematic overcorrection for restriction of range. *Journal of Applied Psychology* 107, 11 (2022), 2040–2068.
- [28] Aaron M. Schmidt and Richard P. DeShon. 2007. What to do? The effects of discrepancies, incentives, and time on dynamic goal prioritization. *Journal of Applied Psychology* 92, 4 (2007), 928–941.
- [29] Frank L. Schmidt and John E. Hunter. 1998. The validity and utility of selection methods in personnel psychology: Practical and theoretical implications of 85 years of research findings. *Psychological Bulletin* 124, 2 (1998), 262–274.
- [30] Frank L. Schmidt, In-Sue Oh, and Jonathan A. Shaffer. 2016. The validity and utility of selection methods in personnel psychology: Practical and theoretical implications of 100 years of research findings. Working Paper.
- [31] Steven E. Scullen, Paul K. Bergey, and Lynda Aiman-Smith. 2005. Forced distribution rating systems and the improvement of workforce potential: A baseline simulation. *Personnel Psychology* 58, 1 (2005), 1–32.
- [32] Steven E. Scullen, Michael K. Mount, and Maynard Goff. 2000. Understanding the latent structure of job performance ratings. *Journal of Applied Psychology* 85, 6 (2000), 956–970.
- [33] Lisa A. Steelman and Lily Wolfeld. 2018. The manager as coach: The role of feedback orientation. *Journal of Business and Psychology* 33 (2018), 41–53.
- [34] Teun van der Weij, Felix Hofstätter, Ollie Jaffe, Samuel F. Brown, and Francis R. Ward. 2024. AI Sandbagging: Language Models can Strategically Underperform on Evaluations. arXiv:2406.07358.
- [35] John Van Maanen and Edgar H. Schein. 1979. Toward a theory of organizational socialization. *Research in Organizational Behavior* 1 (1979), 209–264.
- [36] Jeffrey B. Vancouver and David V. Day. 2005. Industrial and organisation research on self-regulation: From constructs to applications. *Applied Psychology* 54, 2 (2005), 155–185.
- [37] Lei Wang, Chen Ma, Xueyang Feng, Zeyu Zhang, Hao Yang, et al. 2024. A survey on large language model based autonomous agents. *Frontiers of Computer Science* 18, 6 (2024).
- [38] Qingyun Wu, Gagan Bansal, Jieyu Zhang, Yiran Wu, Beibin Li, et al. 2023. AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversation. arXiv:2308.08155.